AI for Economic Research: Dynamic Models, Language, and Agents
Lecture 7.5b: Applications and Validation

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2026-07-07 05:00:38

Lecture 7 in One Map

you are here ▼

7.1 FRAME	7.2 REPRESENT	7.3 ATTEND	7.4 PRETRAIN	**7.5 MEASURE**
text as measurement	tokens $\longrightarrow$ vectors	attention & Transformer	LMs & alignment	limits, apps, validation
what generates this text?	*how do we turn it into numbers?*	*how does context get encoded?*	*where does a pretrained model come from?*	*is my measure valid — and defensible?*

WHY TEXT = MEASUREMENT | HOW THE MACHINE READS TEXT | MEASURE, VALIDATE, DISCLOSE

the spine: $x_i = g(s_i, a_i, c_i) + \varepsilon_i \longrightarrow m(x_i) \longrightarrow y_i = \beta\, m(x_i) + \gamma z_i + u_i$

before 7.1: Lec03–04 gave you neural nets, back-prop & PyTorch — Lec07 teaches them to read

after 7.5: Lec08 RAG (grounded memory) $\rightarrow$ Lec09 agents (grounded action) $\rightarrow$ Lec10 cases

Route: three featured cases → the 2024–26 frontier → embeddings as regression variables → the AI-exposure cautionary tale → the validation workflow → the thesis (part 2/2 of MEASURE, and the close of Lecture 7).

Three Featured Cases

Word2Vec forecasting, TF-IDF exposure, domain encoders — Thread A and B pay off

Three Featured Case Studies

1. **ECB Word2Prices** (Araujo, Bokan, Comazzi & Lenza, 2025)
   - Word2Vec on ECB statements + BVAR for inflation forecasting
2. **Patents and Worker Exposure** (Kogan et al., 2019)
   - TF-IDF + cosine similarity between patents and occupational tasks
3. **Central Bank Sentiment with BERT** (Pfeifer & Marohl, 2023; Ganguli et al., 2024)
   - Fine-tuned RoBERTa for sentiment toward different economic agents
   - PaECTER patent embeddings for innovation dynamics

After the three deep dives: the 2024–26 frontier, then the tools that turn embeddings into regression variables — and the validation discipline that makes any of it publishable.

Case 1 — ECB Word2Prices: Setup

Araujo, Bokan, Comazzi & Lenza (2025), ECB WP No. 3047

- **Question:** can embeddings of ECB communications improve euro-area inflation forecasting?
- **Data:**
  - ECB press conference introductory statements, 2002Q1–2023Q4
  - Target: HICPx (core inflation), monthly $\rightarrow$ quarterly
- **Method:**
  - For each quarter q, retrain Word2Vec on the corpus available *up to* q (no look-ahead)
  - Average word vectors per statement; aggregate to a quarterly text vector
  - Stack text vector with macro variables; fit a Bayesian VAR
  - Forecast 1–4 quarters ahead
- **Why this design?** Re-training quarterly avoids the look-ahead bias that plagues most LLM-based forecast exercises (T2a's caution, obeyed to the letter), while remaining computationally cheap.

Case 1 — Python Skeleton

```
# Quarterly Word2Vec retraining for ECB inflation forecasting
from gensim.models import Word2Vec
from statsmodels.tsa.api import VAR
import numpy as np

forecasts = []
for q in quarters:
    # 1. Build corpus available at quarter q (no look-ahead)
    corpus_q = [s for s in all_statements if s.date <= q]
    # 2. Train Word2Vec on the available corpus
    w2v = Word2Vec(corpus_q, vector_size=100, window=5,
                   sg=0, min_count=2, epochs=20)
    # 3. Average word vectors per statement, then per quarter
    stmt_vec = [np.mean([w2v.wv[w] for w in s if w in w2v.wv], axis=0)
                for s in corpus_q]
    text_q = aggregate_quarterly(stmt_vec, dates)
    # 4. Stack with macro variables and fit a BVAR
    Y = np.column_stack([macro_q, text_q])
    bvar = VAR(Y).fit(maxlags=4)
    # 5. Forecast 1-4 quarters ahead
    forecasts.append(bvar.forecast(Y, steps=4))
```

Case 1 — Results and Lessons

- **Forecast performance:**
  - BVAR with Word2Vec text features outperforms a baseline AR / BVAR without text at horizons of 1–4 quarters
  - Sentiment dictionaries (e.g. Loughran-McDonald) deliver only marginal gains relative to the no-text baseline
  - Heavy LLMs (BERT, fine-tuned) can match or beat Word2Vec but are much more expensive and harder to retrain quarterly — look-ahead bias becomes a real risk
- **Take-aways for empirical economists:**
  - "Lighter is better" if it lets you respect the timing structure of the forecasting exercise
  - Frequently-retrained Word2Vec is a strong baseline before reaching for an LLM
  - Always document: training corpus cutoff at every step, vector dimension, window, retraining frequency
- Replicable framework with zero GPU requirements — ideal for grad-student empirical work.

Case 2 — Patent Text and Worker Exposure

Kogan, Papanikolaou, Schmidt, Seegmiller (2019, 2023)

- **Question:** how does exposure to technological innovation affect workers' earnings and employment?
- **Method:**
  - Compute TF-IDF vectors for U.S. patent texts and for O*NET occupational task descriptions
  - Measure cosine similarity between each occupation and each patent
  - Aggregate up: occupation-level technological exposure
- **Findings:**
  - High-exposure occupations experience earnings declines and reduced employment stability
  - Effects are present for both low- and high-wage workers
- **Why this is in our lecture:** a textbook example of how a *simple* method (TF-IDF + cosine, T2a's toolkit) on a *thoughtfully chosen* corpus can answer a substantive economic question. You don't need transformers to make a contribution, but you do need a clean identification strategy.

Why Generic LLMs Underperform on Financial Text

The vocabulary mismatch. Generic BERT/GPT was trained on Wikipedia, web text, and books — not on disclosure language.

- **Polarity flips.** "Liability," "aggressive," "exposure" are negative in everyday English but neutral or positive in accounting/finance. Loughran & McDonald (2011) is the canonical correction of the older Harvard-IV-4 list.
- **Hedging language.** "May," "could," "intend to" are stop-words in generic NLP but encode legal disclosure stance in 10-K text.
- **Numbers in context.** Subword tokenizers split `1.2345` into digit tokens; without finance fine-tuning the model treats the magnitude as just another vocabulary item.

Two responses: (1) train a domain-specific encoder (next slide), or (2) zero-shot prompt a frontier LLM and validate carefully (Hansen & Kazinnik, 2023).

Case 3a — FinBERT and CentralBankRoBERTa: Two Domain Encoders

	FinBERT (Araci, 2019; Yang et al., 2020)	**CentralBankRoBERTa** (Pfeifer & Marohl, 2023)
Backbone	BERT-base (12 layers, 110M params)	RoBERTa-base, fine-tuned
Pre-train corpus	Reuters TRC2-Financial + 10-K corpora	RoBERTa-base fine-tuned on ∼12k hand-labeled sentences (plus ECB/BIS pseudo-labels)
Task it ships for	3-way sentiment (positive/negative/neutral) on financial sentences	Agent-specific sentiment: stance toward five macro agents (households, firms, financial sector, government, central bank)
Headline result	Beats LSTM/dictionary baselines on Financial PhraseBank	Agent-specific scores predict regional inflation & employment better than "hawkish/dovish"
When to reach for it	Fast firm-level sentiment at scale	Central-bank communication studies

Both are fine-tuned BERTs — the architecture is Act 3; the pretraining/fine-tuning recipe is Act 4 (base → DAPT on domain text → small labeled head). This deck cares only about *using* them: load weights, feed sentences, get scores.

Domain Models vs. Frontier LLMs: No Single Winner

Hansen & Kazinnik (2023): "GPT-4 reads Fedspeak." Zero-shot GPT-4 classifies FOMC sentence stance at expert-level accuracy; beats fine-tuned BERT/RoBERTa; provides human-consistent explanations.

Gambacorta et al. (BIS WP 1215, 2024): CB-LMs. Domain-adapted encoder beats generic BERT on FOMC stance; GPT-4 still wins on complex narrative tasks.

Practical decision rule:

- **Sentence-level sentiment at scale, fixed taxonomy** → FinBERT / CentralBankRoBERTa (cheap, deterministic, citable).
- **Open-ended interpretation, narrative, novel category** → frontier LLM (GPT-4, Claude) with a logged prompt.
- **In doubt** → run both and report agreement (Cohen's κ).

No single model dominates — the validation workflow (later in this deck) handles the choice. And recall T4: the frontier LLM's scores move with its version; the frozen encoder's do not.

Case 3b — PaECTER for Patent Innovation

Ganguli, Lin & Reynolds (2024), NBER WP

- **Goal:** measure innovation dynamics from patent text using transformer-based embeddings.
- **Model: PaECTER** (Patent Embeddings from Citation-Enhanced Transformer Representations)
  - Initialize from BERT-base
  - Fine-tune on patent corpora
  - Use citation pairs as a supervised signal during fine-tuning (related patents pulled together in embedding space)
- **Findings:**
  - Significantly more accurate similarity judgments than TF-IDF or LDA
  - Distinguishes incremental from radical innovations
  - Long-run trend: patents are increasingly diverging in semantic space (broader exploration)
- **Take-away:** fine-tuned transformer embeddings give a richer, citation-aware view of technological evolution than purely text-based or count-based methods.

The 2024–26 Frontier

Central-bank communication and asset pricing move to LLM-scale

Frontier (2024–25): LLMs for Central Bank Communication

- **CB-LMs** (Gambacorta et al., BIS WP 1215, 2024)
  - Domain-adapted encoder retrained on 37,000 central bank papers + 18,000 speeches
  - Outperforms generic BERT on FOMC stance classification
  - But: GPT-4 wins on complex narrative tasks — no single model dominates
- **GPT-4 for Fedspeak** (Hansen & Kazinnik, 2023)
  - GPT-4 classifies FOMC statement sentences at *expert-level* accuracy
  - Outperforms fine-tuned BERT/RoBERTa; provides human-consistent explanations
  - Zero-shot performance collapses domain-adaptation barrier for many tasks
- **GPT-4o on 25,000 FOMC Sentences** (Federal Reserve FEDS Note, December 2024)
  - Topic classification across 2010–2024 minutes at scale
  - Detects "financial stability" discourse spikes in 2020 and 2023 banking crises
  - LLMs reveal *thematic shifts* that sentiment scores miss
- **74,882 Central Bank Documents** (Silva, Moriya & Veyrune, IMF WP 2025/109)
  - 169 central banks, 1884–2025; LLM-coded stance + directional communication index
  - Forecasts future policy rate changes; documents shift to forward-looking language

Frontier (2024–25): LLMs in Asset Pricing

- **Expected Returns and LLMs** (Chen, Kelly & Xiu, NBER WP 31502, 2024)
  - LLM embeddings of financial news ⇒ cross-sectional equity return prediction
  - Substantially outperforms TF-IDF and momentum; gains largest on *negation* and complex narratives — exactly the static-embedding failure cases from T2b
  - Evidence of slow price adjustment to information that only LLM embeddings extract
- **AI Asset Pricing Models** (Kelly, Kuznetsov, Malamud & Xu, NBER WP 33351, 2025)
  - Transformer architecture embedded directly in the stochastic discount factor M_{t+1}
  - Multi-head attention learns cross-asset dependencies from 132 stock characteristics
  - Large reductions in pricing errors vs. previous ML factor models
- **Asset Embeddings** (Gabaix, Koijen, Richmond & Yogo, NBER WP 33651, 2025)
  - Embeddings learned from *investor portfolio holdings* via skip-gram (same math as T2b's Word2Vec)
  - Arithmetic: `tech` + `growth` − `hardware` ≈ `software firms`
  - GPT-4 generates economic narratives for firm clusters in the learned embedding space
- **Key insight:** the same transformer and embedding toolkit that encodes language can encode financial structure — text and quantitative data are no longer separate worlds.

Embeddings as Regression Variables

Similarity, classification — and the standard errors they deserve

Cosine Similarity as an Economic Variable (Thread B)

The most direct embedding-to-regression path: distance between documents.

- $\mathrm{sim}(i,j) = \frac{\mathbf{e}_i^\top \mathbf{e}_j}{\|\mathbf{e}_i\|\,\|\mathbf{e}_j\|}$; cosine *distance* $= 1 - \mathrm{sim}$. Length-invariant by construction — and that matters, because document length tracks firm size and disclosure habits.
- **Statement-change series:** $\delta_t = 1 - \mathrm{sim}(D_{t-1}, D_t)$ across consecutive FOMC statements gives a continuous "how much did the message change" measure — finer than hawk/dove labels. (Similarity tracking of consecutive Fed statements goes back to Acosta & Meade 2015.)
- **Interpretation discipline — two layers of endogeneity:**
  - the *expected* component: strip out what markets already priced, e.g. with high-frequency windows around the release;
  - the *structural* component: the bank may deliberately say more when conditions worsen — δ_t then responds to the economy rather than shocking it. No embedding quality fixes that; only a design can.

The DGP one last time: δ_t is an $m(x)$. Whether it is a *shock* or an *outcome* is an identification claim about a_t and c_t — not a property of the vector space.

Validation Caveats for Econometricians

- **Cosine similarity is not innocent.** Three common confounds inflate similarity between firm/policy texts:
  - **Industry jargon:** firms in the same SIC code share boilerplate language
  - **Templates:** regulatory filings re-use chunks of text verbatim
  - **Length:** longer documents have smoother averages and look more similar
- **Standard tools for robustness:**
  - Permutation tests — shuffle labels within an industry, check if observed similarity exceeds the within-industry baseline
  - Residual embeddings — subtract the industry mean before measuring similarity
  - Bootstrap the entire pipeline (text $\rightarrow$ embedding $\rightarrow$ measurement $\rightarrow$ regression) for honest standard errors (next slide)
- **Dictionary methods are not innocent either.** The Loughran-McDonald (2011) finance-specific dictionary corrects mismatches in the older Harvard IV-4 list (e.g. "liability" is not negative in financial accounting).
- **Always validate** a sample of model outputs against human coding before scaling up.

The Generated-Regressor Problem

Your regressor was estimated — the second stage must know that.

- An embedding similarity, a topic share, an LLM score: all are *estimates*, carrying error from model choice, training data, and aggregation.
- OLS with a noisy constructed regressor ⇒ standard errors too small, t-statistics too big — the classic two-step problem, now wearing NLP clothes.
- Worst when the first stage is trained on little data: embeddings fit on a few hundred FOMC statements carry large estimation error into every downstream δ_t.
- **Fix: bootstrap the whole pipeline** — resample documents, re-embed, re-aggregate, re-run the regression — so first-stage uncertainty propagates into the reported intervals. Report pipeline-bootstrap intervals next to the naive ones.

Rule of thumb: if deleting one year of documents would visibly move your embedding space, then second-stage standard errors without the full-pipeline bootstrap are overconfident.

Zero-Shot Classification: Measuring Concepts with No Training Labels

Describe the class in a sentence; let the geometry do the rest.

- $\hat{c} = \arg\max_{c_k} \operatorname{sim}(\mathbf{e}_{\mathsf{doc}}, \mathbf{e}_{c_k})$: embed the document and each class *description* ("this firm faces material climate-transition risk"), assign by nearest neighbor.
- **Why it matters for economics:** new constructs — climate exposure, AI adoption, supply-chain geopolitical risk — have *no historical labels*. Zero-shot lets you score decades of filings *backwards* with today's concept, extending Hassan-style risk measures to categories nobody coded at the time.
- **Model choice is make-or-break:** raw BERT embeddings are anisotropic — unrelated sentences look similar because they share frequent words. Use sentence-embedding models trained contrastively for similarity (SBERT via `sentence-transformers`, BGE, E5) — T3b's token-vs-document distinction doing real work.
- **Validation ladder — all three steps, in order:**
  1. consistency across paraphrased class descriptions (Fleiss' $\kappa > 0.6$);
  2. accuracy on a hand-coded gold subsample — consistency is *not* correctness: a systematically wrong description can be perfectly consistent;
  3. external validity against a real economic outcome.

A Cautionary Tale

The AI-exposure literature — when the instrument is the thing it measures

The Literature's Question: How Exposed Is Each Job to AI?

- **The measurement problem:** map ∼19,000 O*NET tasks (or ∼900 SOC occupations) to a numeric *AI exposure* score that can be regressed on wages, employment, and county outcomes.
- **Two pre-LLM generations of measures:**
  1. **Felten, Raj, Seamans (2018, 2021)** — AI Occupational Exposure index (AIOE). Link AI benchmark progress (image recognition, translation, ...) to O*NET ability requirements. *Mechanical mapping; human raters set the abilities.*
  2. **Webb (2020)** — text overlap of AI patent abstracts with O*NET task descriptions. *No raters at all — pure TF-IDF.*
- Both treat the measurement instrument as **fixed and inspectable**. A re-run next year produces the same numbers.
- **Economist's framing:** exposure is a *measurement-error* question, not a CS question. The next slide shows what changes when the instrument is itself an LLM.

Three Generations of AI-Exposure Instruments — Compared

Same target (occupation-level exposure score), three very different instruments:

	Felten, Raj, Seamans (2018, 2021)	**Webb (2020)**	**Eloundou et al. (2023)**
Instrument	Hand-curated map: AI benchmark → O*NET ability	TF-IDF cosine: AI patent abstracts → O*NET tasks	Single LLM call: GPT-4 rates each task on 4-level rubric
Cost per refresh	High (human raters, weeks)	Low (re-tokenize, rerun)	Low per call; high per provider switch
Stability across re-runs	Identical (deterministic mapping)	Identical (deterministic)	**Stochastic**; different provider = different score
Validity assumption	Benchmark progress proxies real capability	AI patent text proxies AI capability	The LLM is a calibrated rater
Auditable by economist?	Yes (rater notes)	Yes (read the patent and the task)	Partially (rationale text); not the model weights
Failure mode	Benchmarks lag the frontier	Patent corpus is noisy	Version drift, hallucination, prompt sensitivity

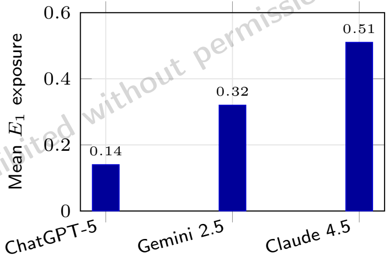
The LLM-as-Rater Turn: Eloundou et al. (2023)

- **Eloundou, Manning, Mishkin, Rock (2023, "GPTs are GPTs")** — the first paper to use **GPT-4 itself** as the rater on the O*NET task list.
- **Rubric (per task):**
  - **E0** — no LLM speedup.
  - **E1** — direct LLM access reduces task time by $\geq 50\%$.
  - **E2** — speedup achievable with LLM-powered software built atop the model.
- **Procedure:** prompt GPT-4 with a task description, ask it to classify, repeat for ∼19k tasks, average within occupation → exposure score.
- Quickly adopted as the *default* exposure measure — Acemoglu (2024), IMF (2024), BIS, and many follow-ups all rebuild on this LLM-rated input.
- **The instrument is now the same kind of model as the subject of the research** — and as T5a established, that instrument is stochastic, hallucination-prone, and version-drifting.

Replication: Yin, Vu, Persico (NBER 35110, 2026)

Design — only the LLM changes:

- Same Eloundou (2023) prompt and task list (18,797 O*NET tasks, ∼900 occupations).
- Three frontier 2026 annotators: **ChatGPT-5**, **Claude 4.5**, **Gemini 2.5**.
- Identical decoding (low temperature, deterministic where supported); identical few-shot examples; identical aggregation to occupation-level E_1 means.
- Plug the resulting exposure scores into the same downstream difference-in-differences regressions used by the labor-AI literature.



Same prompt, same tasks — 3.6× divergence in mean exposure.

- Pairwise agreement: **56.9%**, Cohen's κ as low as **0.36**.
- Individual DiD coefficient varies **2.4**× across annotators.
- County-level estimate **flips sign** (significant negative → insignificant positive).

What 0.36 means (Landis–Koch benchmarks): <0.20 slight, 0.21–0.40 *fair*, 0.41–0.60 moderate, 0.61–0.80 substantial, 0.81–1.00 almost perfect. Publication-grade hand-coding aims for $\kappa > 0.75$

Attitude in the Literature: LLMs Are Not Static Instruments

- **The error is non-classical:** disagreement concentrates on tasks near the rubric boundary — the high-exposure tasks the regressor cares about. Errors correlate with the regressor.
- Yin et al.: *"the risks of treating evolving LLMs as static instruments."* A paper rated with GPT-4 in 2023 cannot be re-run in 2026 and get the same labels.
- **Each T5a limit $\rightarrow$ a measurement consequence:**
  - *Sampling stochasticity* $\rightarrow$ set $\tau = 0$ and report it.
  - *Hallucination / mis-calibration* $\rightarrow$ ensemble across models, report Cohen's κ.
  - *Version drift* $\rightarrow$ pin the snapshot; treat *prompt* + *model* + *seed* as one instrument.
- **Emerging best practice:** pre-register prompt + model + decoding; use multiple frontier annotators and report between-model agreement; audit against a hand-coded gold standard; report inter-annotator sensitivity, not a single point estimate.
- **Takeaway:** the LLM is a measurement instrument — calibrate, version, and disclose.

When NOT to Use an LLM Rater — A Decision Rule

Pearson 0.6 from 50 lines of `re.findall` is a real signal. The LLM rater must clear a meaningful bar to justify its cost.

A defensible workflow before reaching for an LLM:

1. **Hand-code 100–200 task gold standard** (you, an RA, two coders for κ).
2. **Fit a simple baseline** — TF-IDF + logistic regression, or a keyword rule with handful of features. Report κ vs. gold.
3. **Run the LLM in parallel.** Compute κ(LLM, gold) and κ(LLM, baseline).
4. **Decide:**
   - κ(LLM, gold) ≥ 0.75 *and* substantially $>$ κ(baseline, gold) $\Rightarrow$ LLM is validated; use it, disclose model + prompt.
   - κ(LLM, gold) < 0.60 or $\approx$ κ(baseline) $\Rightarrow$ ship the baseline; LLM doesn't earn the version-drift risk.
   - Between 0.60 and 0.75 $\Rightarrow$ ensemble (LLM + baseline + tie-break by hand on disagreements).

The LLM is one instrument among many, not a default. The Yin–Vu–Persico replication shows what you risk by skipping the audition.

The Application Zoo, Organized by Econometric Role

The same text can play three very different roles. Pick the representation to match the role.

Paper	Signal (regressor $m(x_i)$)	Outcome (behavior y)	Instrument / shock
Baker–Bloom–Davis (2016)	Newspaper keyword EPU $\rightarrow$ VAR	—	EPU surprises
Hoberg & Phillips (2016)	10-K TF-IDF cosine $\rightarrow$ product-market similarity	—	—
Hassan et al. (2019)	Bigram firm-level political risk $\rightarrow$ investment regression	—	Quarterly call shocks
Li et al. (2021)	W2V on earnings calls $\rightarrow$ 5 culture dimensions	—	—
Lopez-Lira & Tang (2023)	ChatGPT zero-shot sentiment on headlines $\rightarrow$ next-day returns	—	—
Hansen–McMahon–Prat (2018)	LDA topics	Committee speech change after transparency reform	1993 reform date
Hartley (2025)	LLM-reconstructed EPU (19th c., OCR)	—	War-news shocks

- *Signal column dominates the literature* — it's the easiest role and the loudest result.
- *Outcome* requires a behavior to be *caused* by something else; harder but cleaner identification.
- *Instrument / shock* use needs the exclusion restriction to survive the prompt/fine-tuning recipe — document it.

Validation and Disclosure

The workflow that makes a text measure publishable

A Six-Step Workflow for LLM-Assisted Measurement

1. **Pre-alignment.** Hand-code 100–200 examples; iterate on the prompt or fine-tuning recipe until you reach acceptable agreement.
2. **Full annotation.** Run the model on the entire sample with a frozen prompt and a frozen model version.
3. **Gold-standard audit.** Blind-code 10% of the sample and compute Cohen's κ (publication threshold $\kappa \geq 0.75$).
4. **Category-level reporting.** Don't just report overall accuracy; show the confusion matrix by class. Minority classes are often where models fail silently.
5. **External validity check.** Link the text-derived measure to a real economic outcome (returns, investment, employment) before treating it as a measurement.
6. **Disclosure.** Report prompt, model version, temperature ($\tau = 0$), date, all preprocessing, and provide code. Reproducibility is essential when the model behind the API can change without notice.

Worked Example: Applying the Workflow to FOMC Sentiment

Imagined paper: "GPT-4 sentiment on FOMC statements predicts 10-year yields."

Step	What you do	What goes in the appendix
1. Pre-align	Hand-code 200 sentences hawk/dove/neutral	Prompt v3 (frozen); κ vs. author $= 0.81$
2. Annotate	GPT-4-2024-06-13, $\tau = 0$, full corpus	Cost, runtime, retries, sha256 of inputs
3. Gold audit	Blind-code 10% by RA	$\kappa = 0.78$ overall; 0.62 on "neutral"
4. By-class	Confusion matrix	Hawkish recall 0.91; neutral recall 0.66 (flagged)
5. External validity	Predict Δy_{10} at FOMC dates	$\hat{\beta} = 4.2$ bp per unit, $t = 3.1$
6. Disclose	Prompt + code on Zenodo	DOI, model card, replication run-time

LLM-as-judge connection. A second LLM as a `cross-checker`: same input, different prompt, flag disagreements for human review — the same construction-to-coefficient logic used in the AI-exposure autopsy above, and a preview of Lec09's sub-agent patterns.

Disclosure Norms for AI-Assisted Empirical Work

Minimum reproducibility package.

- **Model identity.** Provider, model name, exact version string, call date.
- **Prompt(s).** Frozen text in a versioned file; not paraphrased in the paper.
- **Sampling parameters.** Temperature, top-p, max tokens, seed if available.
- **Inputs.** sha256 of every input file; document any preprocessing.
- **Outputs.** Raw model outputs preserved *before* any post-processing.
- **Code.** End-to-end script with retry logic and the exact API client version.

The auditable-call triple: $\tau = 0$ + fixed seed + SHA-256 of every input, with raw outputs archived. One reproducible instrument, one line in the appendix.

AEA policy. The October 2024 Data and Code Availability update explicitly covers AI-assisted analysis: report when an LLM was used, for what step, and provide artifacts that let a reader reproduce the result given the same model.

IRB and proprietary text. Survey transcripts, internal memos, and clinical text often cannot be sent to a hosted API. Run an open-weights model locally (T4's menu), or obtain IRB sign-off on the provider's data-handling terms.

The closed-source LLM is the most-changed equipment in your empirical paper. Document it like the dataset it is.

Three Starter Paths for the Macro Researcher

Where to begin, by application — each a full pipeline you can defend.

1. **Central-bank stance.** Domain-adapt an encoder (EconBERT for English; MacBERT + DAPT on PBOC reports for Chinese) $\rightarrow$ extract `[CLS]` embeddings $\rightarrow$ compute δ_t between consecutive statements; benchmark against Romer–Romer narrative shocks as the external anchor; add a LoRA classification head on a small labeled set if you need the hawk/dove *direction*.

2. **Policy uncertainty.** Keep Baker–Bloom–Davis as the transparent, auditable baseline (Thread A's lesson); report a preprocessing specification curve (T2a); test stationarity, and separate *language* breaks from *economic* breaks (T2b) before the VAR.

3. **Nowcasting.** LDA topic shares or embedding principal components as additional factors in the standard nowcast regression — the text block enters exactly like any other factor; but bootstrap the whole text pipeline for the standard errors (this deck).

China parallels ready to run: corporate-culture scores on CNRDS earnings-call text (validate SOE and non-SOE separately — T1); news-based nowcasts on Wind; a PBOC Monetary Policy Report stance series (quarterly since 2001).

The Thesis

What five acts add up to

Identification Precedes Measurement

The lecture's thesis: a better text model shrinks measurement error — it never substitutes for a research design.

- Everything in Acts 2–4 improves $m(x_i)$: less noise, more nuance, more recall. *Nothing* in Acts 2–4 makes $m(x_i)$ exogenous.
- The DGP is the reason: strategic framing a_i and institutional context c_i ride along in every representation, sparse or dense, static or contextual.
- Causal claims still come from designs:
  - **DiD** — Hansen–McMahon–Prat's 1993 reform (T1);
  - **IV** — e.g. exogenous CEO turnover (sudden death, health shocks) shifting communication style, with the exclusion restriction argued, not assumed;
  - **RD**, or **narrative identification** (Romer–Romer) as the external anchor for text-based shock series.
- So the first question of a text project is not "which model?" It is "**which parameter, identified how?**" — the model choice then follows from the role the text plays (T1's three roles).

Identification precedes measurement. Text variables serve the design; they do not replace it.

Lecture 7, Complete

all five acts complete

7.1 FRAME — text as measurement → **7.2 REPRESENT** — tokens → vectors → **7.3 ATTEND** — attention & Transformer → **7.4 PRETRAIN** — LMs & alignment → **7.5 MEASURE** — limits, apps, validation

7.1 FRAME	7.2 REPRESENT	7.3 ATTEND	7.4 PRETRAIN	7.5 MEASURE
what generates this text?	*how do we turn it into numbers?*	*how does context get encoded?*	*where does a pretrained model come from?*	*is my measure valid — and defensible?*

WHY TEXT = MEASUREMENT | HOW THE MACHINE READS TEXT | MEASURE, VALIDATE, DISCLOSE

the spine: $x_i = g(s_i, a_i, c_i) + \varepsilon_i \longrightarrow m(x_i) \longrightarrow y_i = \beta\, m(x_i) + \gamma z_i + u_i$

before 7.1: Lec03–04 gave you neural nets, backprop & PyTorch — Lec07 teaches them to read

after 7.5: Lec08 RAG (grounded memory) → Lec09 agents (grounded action) → Lec10 cases

Nine takeaways in one breath: text is a measurement problem, generated by states, framing, and institutions (1); representations evolved sparse → dense → contextual, each step trading transparency for nuance (2); self-attention is dynamic disambiguation — one vector per occurrence (3); pretrain-then-adapt collapsed the labeled-data requirement (4); alignment made models usable *and* made the version part of the measure (5); five structural limits are architectural facts, not bugs (6); zero-shot classification extends measurement to concepts without labels (7); validation discipline — κ, gold standards, disclosure — is the price of admission (8); and identification precedes measurement (9).

Bridge: From Language Engine to Grounded Systems

Lec07 built the engine; the cure for its parametric weaknesses is grounding.

- **Retrieve** the sources and condition the answer on them — **Lec08 (RAG)**: grounded *memory*. Beats the cutoff and hallucination on *your* corpus; T3b's token-vs-document embeddings are its geometry. (Retraining is too costly and fine-tuning stores style, not facts — retrieval is the practical third path.)
- **Act** through tools that return ground truth — **Lec09 (agents)**: grounded *action*. T4's JSON schemas become tool calls; numbers come from computation, not recollection.
- **The contract either way — cite, then claim:** every factual statement pairs with the retrieved passage or tool output it conditions on. Unpaired claims are parametric memory — auditable only by hand.
- **Lec10** assembles full research cases from these parts.

$$\underbrace{\text{LLM}}_{\text{language engine (Lec07)}} + \underbrace{\text{retrieval}}_{\text{grounded memory (Lec08)}} + \underbrace{\text{tools}}_{\text{grounded action (Lec09)}} \Rightarrow \text{auditable economic measurement}$$

Arc complete: T1 FRAME → T2 REPRESENT → T3 ATTEND → T4 PRETRAIN → **T5 MEASURE**.